Algebra Tutorial

Groups, Homomorphisms, Isomorphisms and the First Isomorphism Theorem

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Problems

1. **Group Operations.** Let G be a group and let $a, b \in G$. Simplify the following expressions:
   (a) $(ab)^{-1}ba$;
   (b) $(aba^{-1})^{-1}$;
   (c) $(ab)^{-1}a^2b$.

2. **Commuting Elements.** Let G be a group and let $a, b \in G$ satisfy $a^3 = e$ and $ab = ba$. Prove that $(ab)^3 = e$.

3. **The Center of a Group.** Let G be a group and define
$$Z(G) = \{z \in G : zg = gz \text{ for all } g \in G\}.$$
Prove that $Z(G)$ is a subgroup of G.

4. **Group Homomorphisms.** Let $\varphi : \mathbb{Z} \to \mathbb{Z}/12\mathbb{Z}$ be defined by
$$\varphi(n) = [5n]_{12}.$$
Show that φ is a group homomorphism, and determine its kernel and image.

5. **Injectivity and Surjectivity.** Let $\varphi : \mathbb{R} \to \mathbb{R}_{>0}$ be defined by
$$\varphi(x) = e^{2x},$$
where the groups are $(\mathbb{R}, +)$ and $(\mathbb{R}_{>0}, \cdot)$. Determine whether φ is a group homomorphism and whether it is an isomorphism.

6. **Kernel and Image.** Let $\varphi : \mathbb{Z} \to \mathbb{Z}$ be defined by
$$\varphi(n) = 6n.$$
Determine $\ker \varphi$ and $\operatorname{im} \varphi$. Is φ injective? Is it surjective?

7. **First Isomorphism Theorem.** Let $\varphi : \mathbb{Z} \to \mathbb{Z}/8\mathbb{Z}$ be the homomorphism given by
$$\varphi(n) = [n]_8.$$
Use the First Isomorphism Theorem to determine $\mathbb{Z}/\ker \varphi$ up to isomorphism.

8. **Group Isomorphisms.** Let $\varphi : \mathbb{Z} \to 4\mathbb{Z}$ be defined by
$$\varphi(n) = 4n.$$
Show that φ is an isomorphism between $(\mathbb{Z}, +)$ and $(4\mathbb{Z}, +)$.

9. **Quotient Groups.** Let G be a group and let $N \trianglelefteq G$. Define
$$\varphi : G \to G/N, \qquad \varphi(g) = gN.$$
Prove that φ is a surjective group homomorphism and determine its kernel. Deduce the corresponding isomorphism using the First Isomorphism Theorem.

10. **First Isomorphism Theorem.** Let
$$\varphi : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}, \qquad \varphi(a, b) = 2a + 3b.$$

(a) Prove that φ is a group homomorphism.

(b) Determine $\ker \varphi$ and $\operatorname{im} \varphi$.

(c) Use the First Isomorphism Theorem to show that

$$(\mathbb{Z} \times \mathbb{Z}) / \ker \varphi \cong \mathbb{Z}.$$